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Socio-Economic Profile and Participation of Local Communities in the Implementation of the Biological Diversity Act, 2002

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Abstract
The effective implementation of the Biological Diversity Act, 2002 is closely linked to the awareness, participation, and traditional knowledge of local communities, who act as primary custodians of biological resources. This study presents an empirical analysis of the socio-economic profile and participatory role of local communities in biodiversity governance across Maharashtra, Dadra and Nagar Haveli, Daman, and Diu. Primary data were collected from 403 respondents through a structured questionnaire and analysed using descriptive statistical techniques. The study examines variables such as age, education, occupation, awareness of the Act, and involvement in conservation activities. The findings reveal that although local communities possess rich traditional ecological knowledge and remain highly dependent on natural resources, their awareness of the legal and institutional framework of the Act is limited. Participation in formal biodiversity governance structures is also relatively low. The study identifies key challenges including low literacy levels, inadequate awareness initiatives, and weak benefit-sharing mechanisms. It concludes that strengthening community engagement, improving awareness, and enhancing institutional support are essential for effective biodiversity conservation and sustainable resource management.

1. Introduction

Role of Local Communities in Biodiversity Governance

The effective implementation of environmental legislation depends not only on institutional frameworks but also on the active involvement of local communities who directly interact with biological resources [1]. Under the Biological Diversity Act, 2002, local communities are recognized as key stakeholders and custodians of traditional knowledge associated with biodiversity. Their practices, developed over generations, contribute significantly to sustainable

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resource management, conservation of ecosystems, and preservation of indigenous knowledge systems [2].

Local communities, particularly in biodiversity-rich regions such as Maharashtra, Dadra and Nagar Haveli, Daman, and Diu, maintain a close relationship with forests, agriculture, and natural resources. Their livelihoods often depend on these resources, making them integral to conservation efforts. Traditional knowledge related to medicinal plants, agricultural practices, water conservation, and ecosystem management plays a vital role in maintaining ecological balance and supporting sustainable development [3].

However, despite their importance, the integration of local communities into formal biodiversity governance remains limited. Awareness of the Biological Diversity Act, 2002, and its institutional mechanisms-such as Biodiversity Management Committees (BMCs) and People's Biodiversity Registers (PBRs)-is often inadequate [4]. Socio-economic constraints, low literacy levels, and lack of institutional support further restrict their participation in decision-making processes and benefit-sharing arrangements [5].

Studies on environmental governance highlight that community participation enhances conservation outcomes, ensures equitable distribution of benefits, and strengthens the legitimacy of policy implementation. In contrast, exclusion or limited involvement of local stakeholders may lead to ineffective enforcement, conflicts, and loss of valuable traditional knowledge. Therefore, understanding the socio-economic characteristics, awareness levels, and participatory role of local communities is essential for evaluating the ground-level effectiveness of biodiversity legislation [6]

In this context, the present study aims to analyse the socio-economic profile and participation of local communities in the implementation of the Biological Diversity Act, 2002. By examining factors such as education, occupation, awareness, and engagement in conservation activities, the study seeks to identify gaps and provide insights for improving community-based biodiversity governance [8]

2. Research Methodology

This study adopts a descriptive and empirical research methodology to examine the socio-economic profile and participation of local communities in the implementation of the Biological Diversity Act, 2002. The methodology is designed to analyse the level of awareness, traditional

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knowledge practices, and involvement of local communities in biodiversity conservation and governance [9-20].

2.1 Research Design

The study follows a descriptive research design, as it aims to systematically describe and analyse the existing conditions related to awareness, participation, and socio-economic characteristics of local communities without manipulating any variables. This design is appropriate for understanding ground-level realities of biodiversity governance.

2.2 Source of Data

The research is primarily based on primary data collected from local community members through a structured questionnaire. Secondary data from government reports, policy documents, research articles, and academic literature were also used to support analysis and interpretation.

2.3 Study Area

The study was conducted across four regions: Maharashtra, Dadra and Nagar Haveli, Daman, and Diu. These areas were selected due to their ecological diversity, presence of indigenous communities, and relevance in biodiversity conservation.

2.4 Sample Size and Sampling Technique

A total of 403 respondents from local communities constituted the sample for the study. The respondents were selected using a stratified random sampling technique to ensure representation of different socio-economic groups and geographical areas.

2.5 Data Collection Tool

A structured questionnaire was used as the primary data collection instrument. The questionnaire included variables such as age, education level, occupation, awareness of the Biological Diversity Act, 2002, involvement in biodiversity conservation activities, and knowledge of traditional practices. The tool was designed to ensure clarity, simplicity, and relevance to the research objectives.

2.6 Data Analysis Techniques

The collected data were coded and analysed using descriptive statistical methods. Frequency and percentage analysis were applied to identify patterns in awareness, participation, and socio-economic characteristics. The results were presented in tabular and graphical forms for clarity and interpretation.

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2.7 Ethical Considerations

Participation in the study was voluntary. Respondents were informed about the purpose of the research, and confidentiality of their responses was strictly maintained. No personal identifiers were used in data analysis or reporting.

2.8 Limitations of the Methodology

The study is limited to selected regions and may not fully represent all local communities across India. The analysis is confined to socio-economic and awareness variables and does not directly measure conservation outcomes. Additionally, responses are based on self-reporting, which may involve certain biases.

3. Results and Discussion

This section presents the analysis and interpretation of data collected from local community respondents regarding their socio-economic characteristics and participation in biodiversity conservation.

3.1 Age-wise Distribution of Local Communities

Table 1: Age-wise Distribution of **Local Community Members**

Age Group	**Frequency**	**Percentage (%)**
18–35 Years	120	30%
36–60 Years	210	52%
61 Years & Above	73	18%
Total	**403**	**100%**

Interpretation

The above table shows the age-wise distribution of 403 local community respondents. A majority of respondents (52%) fall within the age group of 36–60 years, indicating that middle-aged individuals constitute the largest segment of the population. This group represents the active workforce and plays a significant role in livelihood activities such as agriculture, forest resource utilization, and biodiversity conservation.

The younger age group (18–35 years) accounts for 30% of respondents. This segment reflects the emerging generation that has the potential to adopt new conservation practices and participate in awareness programs related to the Biological Diversity Act, 2002.

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The senior age group (61 years and above) comprises 18% of respondents. Although smaller in proportion, this group is highly important as it represents the custodians of traditional knowledge, including indigenous practices related to medicinal plants, agriculture, and environmental management.

Overall, the age distribution reflects a balanced composition of experienced individuals, active participants, and knowledge holders. This diversity is beneficial for effective biodiversity governance; however, the relatively lower representation of elderly respondents highlights the need for systematic documentation and preservation of traditional knowledge.

3.2 Education Level of Local Communities

Table 2: Education Level of Local Community Members

Education Level	Frequency	Percentage (%)
Illiterate	40	10%
Primary Schooling	52	13%
Higher Secondary / Diploma	142	35%
Graduation	113	28%
Post Graduation	56	14%
Total	**403**	**100%**

Interpretation

The above table presents the educational profile of 403 local community respondents. The largest proportion of respondents (35%) have attained higher secondary education or a diploma, followed by 28% who are graduates. Additionally, 14% of respondents have completed post-graduation, indicating the presence of a reasonably educated segment within the community.

At the lower end of the educational spectrum, 13% of respondents have completed only primary schooling, while 10% are illiterate. This indicates that although a small section of the population still lacks basic education, the majority of respondents (approximately 77%) have attained at least secondary-level education or higher

The relatively higher level of education among local community members is a positive indicator for the implementation of the Biological Diversity Act, 2002, as education enhances awareness, understanding of legal provisions, and participation in conservation initiatives. Educated individuals are more likely to engage with institutional mechanisms such as Biodiversity Management Committees and contribute to sustainable practices.

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However, the presence of illiterate and less-educated respondents highlights the need for targeted awareness programs, training, and capacity-building initiatives in simple and accessible formats. Bridging this educational gap is essential to ensure inclusive participation and effective implementation of biodiversity policies at the grassroots level.

3.3 Occupation-wise Distribution of Local Communities

Table 3: Occupation of Local Community Members

Occupation	Frequency	Percentage (%)
Self-Employed	133	33%
Farmers	107	26%
Wage-Based Workers	96	24%
Private Employees	37	9%
Businessmen	30	7%
Total	**403**	**100%**

Interpretation

The above table illustrates the occupational distribution of local community respondents. A significant proportion of respondents are self-employed (33%), followed by farmers (26%) and wage-based workers (24%). Together, these three categories constitute approximately 83% of the total respondents, indicating a strong dependence on primary and informal economic activities

The dominance of self-employment and agriculture reflects the rural and semi-rural nature of the study areas, where livelihoods are closely linked to natural resources such as land, forests, and water. Wage-based workers also form a substantial portion, suggesting reliance on daily or seasonal labour, often associated with agriculture, forestry, and allied sectors.

In contrast, a relatively smaller percentage of respondents are engaged as private employees (9%) and businessmen (7%). This indicates limited penetration of formal employment and organized business activities within local communities.

The occupational pattern clearly highlights that the majority of the population is directly dependent on biodiversity for their livelihood. This dependence makes local communities' key stakeholders in biodiversity conservation and sustainable resource management. Their involvement is essential for the successful implementation of the Biological Diversity Act, 2002. However, the prevalence of informal and low-income occupations may also limit awareness and participation in formal governance structures. Therefore, targeted livelihood support, awareness programs, and community-based initiatives are necessary to enhance both economic stability and active participation in biodiversity conservation.

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3.4 Awareness of Biological Diversity Act, 2002

Table 4: Awareness of the Objectives and Provisions of the Biological Diversity Act, 2002

Response Category	Frequency	Percentage (%)
Strongly Disagree	23	6%
Disagree	98	24%
Neutral	185	46%
Agree	70	17%
Strongly Agree	27	7%
Total	**403**	**100%**

Interpretation

The above table reflects the level of awareness among local community members regarding the objectives and provisions of the Biological Diversity Act, 2002. A significant proportion of respondents (46%) reported a neutral stance, indicating uncertainty or lack of clear understanding of the Act. Additionally, 24% disagreed and 6% strongly disagreed, suggesting that nearly 30% of respondents lack awareness about the Act.

On the other hand, only 17% agreed and 7% strongly agreed that they are aware of the Act, indicating that a relatively small proportion of the community possesses adequate knowledge.

Overall, the findings clearly highlight that awareness levels among local communities are low to moderate, with a majority either lacking knowledge or being uncertain. This lack of awareness acts as a major barrier to effective implementation of the Act, as community participation and compliance depend heavily on understanding legal provisions.

The results emphasize the urgent need for awareness campaigns, training programs, and community-level outreach initiatives to enhance knowledge about biodiversity laws and empower local communities to actively participate in conservation efforts.

3.5 Participation in Biodiversity Conservation

Table 5: Perception on Improvement in Community Participation in Biodiversity Conservation

Response Category	Frequency	Percentage (%)
Strongly Disagree	26	6%
Disagree	87	22%
Neutral	170	42%
Agree	98	24%
Strongly Agree	22	5%
Total	**403**	**100%**

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Interpretation

The above table presents the perception of local community members regarding the role of the Biological Diversity Act, 2002 in improving participation in biodiversity conservation. A large proportion of respondents (42%) remained neutral, indicating uncertainty or lack of direct experience with the impact of the Act on community participation.

Further, 22% of respondents disagreed and 6% strongly disagreed that the Act has improved participation, suggesting that a considerable segment of the population does not perceive significant positive change.

In contrast, 24% agreed and 5% strongly agreed that the Act has enhanced community participation. Although this reflects some level of positive impact, the proportion remains relatively low.

The findings indicate that community participation in biodiversity conservation is still limited and not strongly influenced by the Act. The high percentage of neutral responses suggests a gap between policy implementation and ground-level impact.

This highlights the need for strengthening institutional mechanisms, promoting active involvement of local communities, and increasing awareness about participatory governance structures such as Biodiversity Management Committees (BMCs). Enhancing community engagement is essential for achieving effective biodiversity conservation and sustainable development.

3.7 Challenges Faced by Local Communities

The below table presents the major challenges faced by local community members in the implementation of the Biological Diversity Act, 2002. The findings indicate that most of the challenges are perceived as moderate to highly significant, reflecting serious gaps in effective implementation at the grassroots level.

One of the most prominent challenges is the limited awareness of the Act, with 31% of respondents considering it very challenging and 8% highly challenging. This highlights a fundamental issue, as awareness is essential for participation and compliance.

Another critical issue is the lack of financial support for biodiversity conservation, where a similar proportion of respondents identified funding constraints as a major barrier. This affects the ability of communities to actively engage in conservation activities.

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Table 6: Challenges Faced by Local Community Members in the Implementation of the Biological Diversity Act, 2002

Sr. No	Challenges	Moderately Challenging (%)	Very Challenging (%)	Highly Challenging (%)
1	Limited awareness of the Act's objectives and provisions	36%	31%	8%
2	Lack of funding for biodiversity conservation initiatives	36%	31%	8%
3	Insufficient community engagement	37%	32%	7%
4	Difficulty in enforcing legal provisions due to lack of resources	34%	30%	8%
5	Risk of exploitation of traditional knowledge by external entities	36%	32%	8%
6	Lack of inter-departmental coordination	35%	30%	9%
7	Limited technical expertise and training	36%	31%	7%
8	Lack of reliable data on biodiversity and traditional knowledge	35%	32%	7%
9	Complex legal framework and procedural delays	35%	30%	8%
10	Resistance due to lack of incentives or fear of losing traditional rights	35%	30%	7%

The study also reveals insufficient community engagement, indicating that local communities are not adequately involved in decision-making processes. This is further compounded by the lack of coordination among departments, which creates administrative inefficiencies.

Additionally, respondents reported limited technical expertise and training, suggesting that communities lack the necessary skills to implement conservation practices effectively. The absence of reliable data on biodiversity and traditional knowledge further restricts informed decision-making.

Concerns regarding the exploitation of traditional knowledge by external entities were also highlighted, reflecting fears of biopiracy and lack of benefit-sharing mechanisms. Moreover, the complex legal framework and procedural delays act as significant barriers, making it difficult for communities to understand and engage with the Act.

Finally, resistance from stakeholders, often due to lack of incentives or fear of losing traditional practices, further limits participation.

Overall, the findings indicate that while local communities are central to biodiversity conservation, they face multiple structural, economic, and institutional challenges. Addressing these issues through awareness programs, financial support, capacity building, and simplified legal mechanisms is essential for improving the effectiveness of the Biological Diversity Act, 2002.

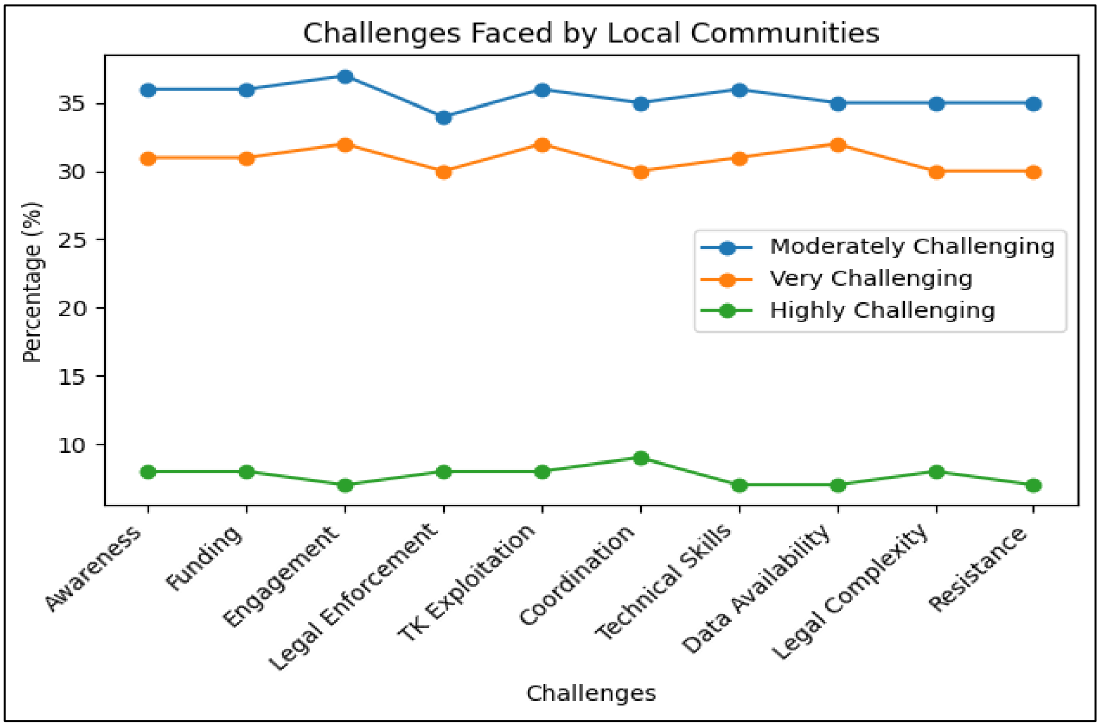

Figure 1: Challenges Faced by Local Community Members

4. Conclusion

The present study examined the socio-economic profile, awareness, participation, and challenges faced by local communities in the implementation of the Biological Diversity Act, 2002 in selected regions of Maharashtra, Dadra and Nagar Haveli, Daman, and Diu. The findings clearly demonstrate that local communities play a crucial role as custodians of biodiversity and traditional knowledge; however, their potential remains underutilized due to several structural and institutional limitations.

The demographic analysis indicates a balanced age distribution with a dominant presence of middle-aged individuals, supported by younger populations and a smaller proportion of elderly knowledge holders. The educational profile suggests that a significant portion of respondents

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possesses at least secondary-level education, which provides a strong foundation for awareness and participation. At the same time, the occupational structure highlights a heavy dependence on agriculture, self-employment, and wage-based work, reflecting a close relationship between livelihoods and natural resources.

Despite this strong connection with biodiversity, the study reveals that awareness of the Biological Diversity Act, 2002 among local communities is limited. A large proportion of respondents either lack clear understanding or remain uncertain about the objectives, provisions, and legal implications of the Act. This gap in awareness directly affects their level of participation in formal biodiversity governance mechanisms such as Biodiversity Management Committees.

The findings also indicate that community participation in biodiversity conservation remains moderate to low. Although some respondents acknowledge the positive impact of the Act, a substantial number express uncertainty or dissatisfaction, suggesting that the benefits of the Act have not been fully realized at the grassroots level.

Furthermore, the study identifies several critical challenges, including lack of awareness, insufficient funding, weak institutional coordination, limited technical expertise, and concerns regarding the exploitation of traditional knowledge. The complexity of the legal framework and procedural delays further hinder effective implementation. These challenges collectively restrict the active involvement of local communities in biodiversity governance.

In conclusion, the study emphasizes that the success of the Biological Diversity Act, 2002 depends not only on institutional mechanisms but also on the meaningful inclusion of local communities. Strengthening awareness, enhancing capacity-building initiatives, simplifying legal processes, and ensuring equitable benefit-sharing mechanisms are essential to improve participation and implementation outcomes. Integrating local knowledge systems with formal governance structures can significantly contribute to sustainable biodiversity conservation and long-term environmental protection.

Conflict of Interest: The author(s) declare that there is no conflict of interest regarding the publication of this paper

References

1. Sharma, D., Singh, P., & Punhani, A. (2024). Sugarcane Diseases Detection Using Optimized Convolutional Neural Network with Enhanced Environmental Adaptation Method. International Journal of

Experimental Research and Review, 41(Spl Vol), 55-71. https://doi.org/10.52756/ijerr.2024.v41spl.005
2. Banerjee, S., Mitra, S., Velhal, M., Desmukh, V., & Ghosh, B. (2021). Impact of agrochemicals on the environment and human health: The concerns and remedies. *Int. J. Exp. Res. Rev.*, 26, 125-140. https://doi.org/10.52756/ijerr.2021.v26.010
3. Subhashree Subhasmita et al. "An Extensive Study: Preparation of People's Biodiversity Register (PBR) in Malkangiri District, Odisha." *Bioscience Biotechnology Research Communications* (2021). https://doi.org/10.21786/bbrc/14.9.1
4. Subhashree Subhasmita et al. "An Extensive Study: Preparation of People's Biodiversity Register (PBR) in Malkangiri, Odisha." *Bioscience Biotechnology Research Communications* (2021). https://doi.org/10.21786/bbrc/14.9.58
5. Mangana B. Rampheri et al. "Local community attitudes and perceptions towards benefits and challenges associated with biodiversity conservation in Blouberg Nature Reserve, South Africa." *African Journal of Ecology* (2022). https://doi.org/10.1111/aje.12989
6. Bhartiya Krishi et al. "Protecting India's Biodiversity: The Role of the Biological Diversity Act 2002: A Review." *Bhartiya Krishi Anusandhan Patrika* (2025). https://doi.org/10.18805/bkap788
7. De, M., Pahari, G., & Das, R. (2019). Creating Urban Green Spaces (UGS) in Educational Institutions: A pilot project in Gurudas College, Kolkata-700054, West Bengal, India *Int. J. Exp. Res. Rev.*, 19, 22-30. https://doi.org/10.52756/ijerr.2019.v19.003
8. Yuling Zhang et al. "How important is community participation to eco-environmental conservation in protected areas? From the perspective of predicting locals' pro-environmental behaviours.." *The Science of the total environment*, 739 (2020): 139889. https://doi.org/10.1016/j.scitotenv.2020.139889
9. Hussein Gena Koricha et al. "Investigated the role of community based approaches for biodiversity conservation and socio-economic development in Bale Mountains National Park, Southeast Ethiopia." *Scientific Reports*, 14 (2024). https://doi.org/10.1038/s41598-024-60177-5
10. Amritpal Singh Kaleka et al. "Community Conservation." *Endangered Plants* (2020). https://doi.org/10.5772/intechopen.94557
11. C. Ward et al. "Protected area co-management and perceived livelihood impacts.." *Journal of environmental management*, 228 (2018): 1-12. https://doi.org/10.1016/j.jenvman.2018.09.018
12. De, M., Pahari, G., & Das, R. (2019). Creating Urban Green Spaces (UGS) in Educational Institutions: A pilot project in Gurudas College, Kolkata-700054, West Bengal, India *Int. J. Exp. Res. Rev.*, 19, 22-30. https://doi.org/10.52756/ijerr.2019.v19.003
13. Subhashree Subhasmita et al. "An Extensive Study: Preparation of People's Biodiversity Register (PBR) in Malkangiri District, Odisha." *Bioscience Biotechnology Research Communications* (2021). https://doi.org/10.21786/bbrc/14.9.1

14. Subhashree Subhasmita et al. "An Extensive Study: Preparation of People's Biodiversity Register (PBR) in Malkangiri, Odisha." *Bioscience Biotechnology Research Communications* (2021). https://doi.org/10.21786/bbrc/14.9.58
15. Mangana B. Rampheri et al. "Local community attitudes and perceptions towards benefits and challenges associated with biodiversity conservation in Blouberg Nature Reserve, South Africa." *African Journal of Ecology* (2022). https://doi.org/10.1111/aje.12989
16. Bhartiya Krishi et al. "Protecting India's Biodiversity: The Role of the Biological Diversity Act 2002: A Review." *Bhartiya Krishi Anusandhan Patrika* (2025). https://doi.org/10.18805/bkap788
17. Yuling Zhang et al. "How important is community participation to eco-environmental conservation in protected areas? From the perspective of predicting locals' pro-environmental behaviours.." *The Science of the total environment*, 739 (2020): 139889. https://doi.org/10.1016/j.scitotenv.2020.139889
18. Hussein Gena Koricha et al. "Investigated the role of community based approaches for biodiversity conservation and socio-economic development in Bale Mountains National Park, Southeast Ethiopia." *Scientific Reports*, 14 (2024). https://doi.org/10.1038/s41598-024-60177-5
19. Amritpal Singh Kaleka et al. "Community Conservation." *Endangered Plants* (2020). https://doi.org/10.5772/intechopen.94557
20. C. Ward et al. "Protected area co-management and perceived livelihood impacts.." *Journal of environmental management*, 228 (2018): 1-12. https://doi.org/10.1016/j.jenvman.2018.09.018

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